Database Principles

CCS 3400

Data Definition Language and Data Control Language

Learning Objectives

- Explain SQL and its categories
- Define and modify a database using DDL
- Apply constraints to enforce integrity
- Manage access with DCL (GRANT/REVOKE) to access database or particular object within the database.

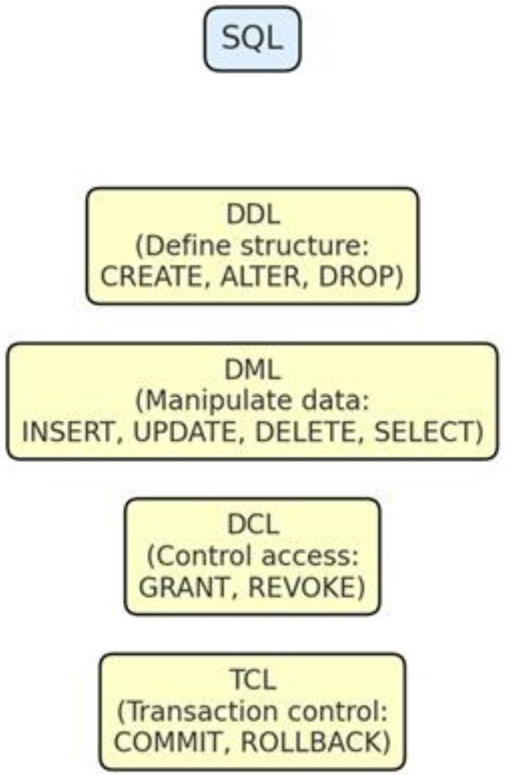
What is SQL?

- SQL or Structured Query Language is an English-like language used to create and manipulate databases.
- Three types of statements:
  - **Data Definition Language** (DDL)
  - **Data Manipulation Language** (DML)
  - **Data Control Language** (DCL)
- Data Definition Language creates and manipulates the structure that Data Manipulation Language fills. Data Control Language determines who is allowed to what within the database

Structured Query Language (SQL)

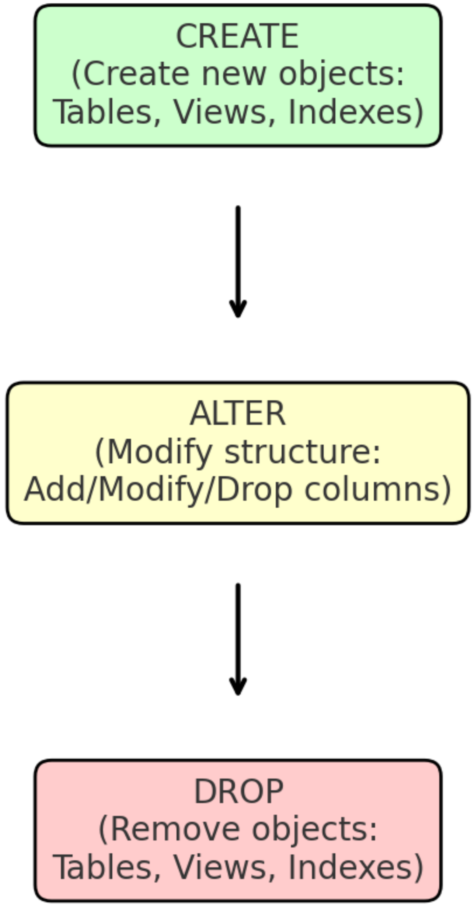
- Structured Query Language (SQL):
  - DDL: Defines structure (CREATE, ALTER, DROP)
  - DML: Manipulates data (INSERT, UPDATE, DELETE, SELECT)
  - DCL: Controls access (GRANT, REVOKE)
  - TCL: Transaction control (COMMIT, ROLLBACK, SAVEPOINT)

- Analogy:
  - DDL = Architect | DML = Builder | DCL = Gatekeeper | TCL = Manager



Data Definition Language (DDL)

- Data Definition Language (DDL) is the part of SQL used to **define and manage the structure of a database**.
- It deals with creating, modifying, and deleting database objects such as **tables, schemas, views, and indexes**.
- **Key Points:**
  - **Defines the schema** (the "blueprint" of the database).
  - Common commands:
    - **CREATE** → build new objects (tables, indexes, views).
    - **ALTER** → change the structure (add/modify/drop columns).
    - **DROP** → remove objects permanently.
  - Includes **constraints** (e.g., PRIMARY KEY, FOREIGN KEY, UNIQUE, CHECK, NOT NULL) to maintain data integrity.
- 👉 In simple words: **DDL is like the architect's design** – it sets up the framework where data will live.



Data Manipulation Language (DML)

- **Data Manipulation Language (DML)** is the part of SQL that deals with **working with the actual data** stored in a database.

🔹 Key Points

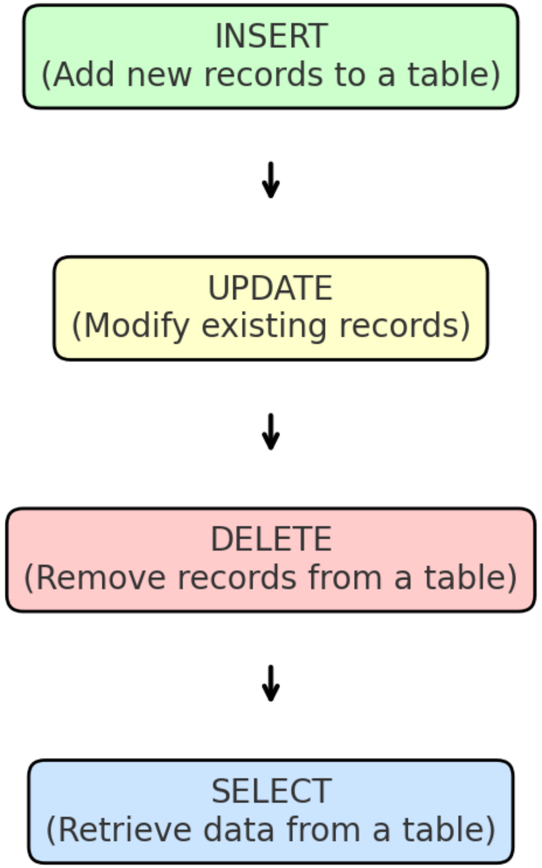
- **Purpose**: To insert, update, delete, and retrieve data from tables.
- **Common DML Commands**:

  INSERT → Add new data into a table.
  UPDATE → Modify existing records.
  DELETE → Remove records.
  SELECT → Retrieve data.

🔹 Simple Analogy

If **DDL is the architect** (designing the building), then **DML is the people using the building** (adding, changing, or removing items inside).

👉 In short: **DML is all about handling the data inside the database tables.**



Example…

```
CREATE TABLE room(
    roomID              number,
    bldg                char(1)             CHECK (bldg IN ('A','B')),
    roomNo    varchar2(10),
    maxCapacity         number,
    style               varchar2(15)        CHECK (style IN ('LECTURE',
                                  'LECTURE/LAB', 'LAB','OFFICE')),
    CONSTRAINT room_pk PRIMARY KEY (roomID));
```

Example…

```
CREATE TABLE faculty(
        facultyID  number,
        lname               varchar2(30)   NOT NULL,
        fname               varchar2(20)   NOT NULL,
        dept                varchar2(5),
        officeID            number,
        phone               varchar2(15),
        email               varchar2(75)   UNIQUE,
        rank                char(4)          CHECK (rank IN ('INST','ASOC',
                                          'ASST','FULL','SENR')),
  CONSTRAINT faculty_pk PRIMARY KEY (facultyID),
  CONSTRAINT faculty_fk FOREIGN KEY (officeID) REFERENCES
   room(roomID));
```

Table with Primary Key Only

```
CREATE TABLE room(
        roomID  number,
        bldg    char(1) CHECK (bldg IN ('A','B')),
        roomNo varchar2(10),
        maxCapacity     number,
        style   varchar2(15)  CHECK (style IN
    ('LECTURE','LECTURE/LAB','LAB','OFFICE')),
  CONSTRAINT room_pk PRIMARY KEY (roomID));
```

Table with Foreign Key

```
CREATE TABLE faculty(
     facultyID     number,
     lname         varchar2(30)     NOT NULL,
     fname         varchar2(20)     NOT NULL,
     dept          varchar2(5),
     officeID      number,
     phone    varchar2(15),
     email         varchar2(75)     UNIQUE,
     rank          char(4)  CHECK (rank IN ('INST',
                 'ASOC','ASST','FULL','SENR')),
 CONSTRAINT faculty_pk PRIMARY KEY (facultyID),
 CONSTRAINT faculty_fk FOREIGN KEY (officeID) REFERENCES room(roomID));
```

Table with Compound Primary Key

```
CREATE TABLE participation(
    eventID  number,
    memberID        number,
    CONSTRAINT participation_pk primary key (eventID, memberID),
    CONSTRAINT participation_event_fk foreign key (eventID)
      references event(eventID),
    CONSTRAINT participation_member_fk foreign key (memberID)
      references member(memberID));
```

DDL – Altering table

Use the Alter and Add/Modify keywords and specify:

- table name
- new or existing field name(s)
- field type

```
ALTER TABLE invoice
ADD(sent_dt date);

ALTER TABLE invoice
MODIFY (invoice_nbr varchar2(5));
```

DDL – Removing a Table

Use the Drop keyword and specify:

- table name

```
DROP TABLE invoice;
```

- If this table is referenced by a foreign key, use the **cascade constraints clause**

```
DROP TABLE invoice cascade constraints;
```

Indexes

- Conceptually similar to book index
- Increases data retrieval efficiency
- Automatically assigns record numbers
- Used by DBMS, not by users
- Fields on which index built called *Index Key*
- Have a sorted order
- Can guarantee uniqueness
- Can include one or more fields

Indexes – Syntax

```
CREATE UNIQUE INDEX cust_num_ind ON
 customer(Name);

CREATE INDEX CustomerName ON Customer
  (CustomerNum);

CREATE INDEX CreditLimitRep_ind ON
 Customer(CreditLimit, RepNum);

DROP INDEX RepBal;
```

Customer Table with Record Numbers Figure 4.10

RecordNum	CustomerNum	CustomerName	...	Balance	CreditLimit	RepNum
1	148	Al's Appliance and Sport	...	$6,550.00	$7,500.00	20
2	282	Brookings Direct	...	$431.50	$10,000.00	35
3	356	Ferguson's	...	$5,785.00	$7,500.00	65
4	408	The Everything Shop	...	$5,285.25	$5,000.00	35
5	462	Bargains Galore	...	$3,412.00	$10,000.00	65
6	524	Kline's	...	$12,762.00	$15,000.00	20
7	608	Johnson's Department Store	...	$2,106.00	$10,000.00	65
8	687	Lee's Sport and Appliance	...	$2,851.00	$5,000.00	35
9	725	Deerfield's Four Seasons	...	$248.00	$7,500.00	35
10	842	All Season	...	$8,221.00	$7,500.00	20

Customer Table Index on CustomerNum Figure 4.11

CustomerNum	RecordNum
148	1
282	2
356	3
408	4
462	5
524	6
608	7
687	8
725	9
842	10

Table Indexes on CreditLimit, RepNum Figure 4.12

CreditLimit	RecordNum
$5,000	4, 8
$7,500	1, 3, 9, 10
$10,000	2, 5, 7
$15,000	6

RepNum	RecordNum
20	1, 6, 10
35	2, 4, 8, 9
65	3, 5, 7

Indexes

PROs

- Faster/more efficient data retrieval

CONs

- Requires additional space
- Increased overhead
- Slower writes

Data Control Language (DCL)

Used to control database privileges

- Grant
- Revoke

DCL – Grant Privilege

Use the Grant keywords and specify:

- Privilege
  {all, select, insert, update, delete, index, alter}
- existing field name(s)
- username

```
GRANT ALL on employee to salmi;
GRANT SELECT on employee to public;
GRANT SELECT, INSERT, DELETE on
employee to ahmad;
```

DCL – Revoke Privilege

Use the Revoke keywords and specify:

- Privilege
  {all, select, insert, update, delete, index, alter}
- existing field name(s)
- username

```
REVOKE ALL on employee from salmi;
REVOKE SELECT on employee from public;
REVOKE UPDATE, DELETE on employee from
ahmad;
```

Best Practices

- Use roles to simplify user management
- Apply constraints for data integrity
- Document schema changes
- Grant only what is necessary

Summary Diagram of DDL vs DML vs DCL roles

DDL
(Data Definition Language)

- Defines structure
- CREATE, ALTER, DROP
- Architect role

DCL
(Data Control Language)

- Controls access
- GRANT, REVOKE
- Gatekeeper role

DML
(Data Manipulation Language)

- Manipulates data
- INSERT, UPDATE, DELETE, SELECT
- Builder role

Summary

- DDL (Data Definition Language) – used to create, change, and delete database structures (tables, columns, constraints).
- Constraints – rules like PRIMARY KEY, FOREIGN KEY, UNIQUE, NOT NULL, and CHECK help keep data accurate and consistent.
- DCL (Data Control Language) – manages permissions in the database using GRANT and REVOKE.
- Best practice – give only the access that is needed (principle of least privilege).
- Together, DDL = structure and DCL = security, making the database reliable and safe.

Q&A

1. Why is DROP TABLE risky?
2. How does CHECK constraint improve data quality?
3. Difference between granting to user vs role?

End of Chapter